

A

Internship Report on

Mental Health assessment with AI

Submitted in partial fulfillment of the requirements for the
award of Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

By

Aakanksha Laxmi(160622733122)

Under the Guidance of

Mrs.Shaheda Begum

Assistant professor

Department of Computer Science and Engineering



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN
(AUTONOMOUS)**

(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)
Hyderabad - 500 001, Telangana, India

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(AUTONOMOUS)

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Hyderabad - 500 001, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the Internship report entitled, “**Mental Health assessment with AI**”, done by **Aakanksha Laxmi(160622733122)**, Submitting to the Department of Computer Science and Engineering, **STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**, in partial fulfillment of the requirements for the Degree of **BACHELOR OF ENGINEERING** in **Computer Science and Engineering**, (at-company name) during the year 2024-25. It is certified that he/she has completed the project satisfactorily.

Signature of Supervisor / Faculty Coordinator

Mrs.Shaheda Begum
Assistant professor

Signature of Head of the Department

Dr. R. Madana Mohana
Professor & Head

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Hyderabad - 500 001, Telangana, India
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



DECLARATION

I hereby declare that the work described in this report entitled "**Mental Health assessment with AI**" which is being submitted by us in partial fulfillment for the award of **BACHELOR OF ENGINEERING** in the Department of Computer Science and Engineering, Stanley College of Engineering & Technology for Women to the Osmania University Hyderabad.

The work is original and has not been submitted for any Degree or Diploma of this or any other university.

Signature of the Student

Aakanksha Laxmi

(160622733122)

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INTERNSHIP - II DETAILS

Course Code:SPW611CS

Credits: 1

Evaluation: CIE - 50 Marks

Title: Mental Health assessment with AI

Name of the Student:Aakanksha Laxmi

Roll Number:160622733122

Batch / Section: 2024-2025/B

Name of the Organization: Defence Research and Development Laboratory

Duration of Internship: From 28/04/2025 To 28/05/2025

Mode (Onsite / Online / Hybrid):Hybrid

Industry Supervisor: Y.Harinath, DASQA, DRDL.

Faculty Coordinator / Supervisor: Mrs.Shaheda Begum

Organization Profile

• Overview of the organization

The Defence Research and Development Laboratory (DRDL) is one of the premier research laboratories under the Defence Research and Development Organisation (DRDO), Ministry of Defence, Government of India. It is located in Hyderabad, Telangana, and is primarily engaged in the design, development, and production of missile systems and related technologies for the Indian Armed Forces.

Established in the early 1960s, DRDL was created with the vision of achieving self-reliance in guided missile technology. Over the decades, it has played a pivotal role in developing a wide range of strategic and tactical missile systems, contributing significantly to India's defence capabilities. DRDL focuses on advanced technologies such as aerodynamics, propulsion, guidance and control, warhead design, and system integration.

The laboratory has been instrumental in the successful development of missile programs such as Prithvi, Akash, Nag, Agni, and K-15, among others. It collaborates closely with other DRDO laboratories, academic institutions, and industrial partners to ensure continuous technological advancement and innovation.

In addition to research and development, DRDL also emphasizes testing, validation, and simulation-based design, ensuring that the systems developed meet the highest standards of quality, reliability, and performance. Its multidisciplinary teams of scientists, engineers, and technicians work together to meet the evolving strategic and tactical requirements of India's defence forces.

Vision, Mission, and Key Areas of Operation:

Vision:

To achieve excellence in the field of missile technology and contribute to India's self-reliance in defence systems.

Mission:

To design, develop, and deliver state-of-the-art missile systems and technologies to meet the present and future requirements of the Indian Armed Forces.

Key Focus Areas:

- Design and development of strategic and tactical missile systems
- Research in propulsion systems, control mechanisms, and aerodynamics
- Integration and testing of missile subsystems
- Collaboration with industry and academic institutions for technology transfer

Organizational Structure:

The Defence Research and Development Laboratory (DRDL) operates under the Defence Research and Development Organisation (DRDO), which functions under the Ministry of Defence, Government of India. DRDL follows a structured hierarchical organization designed to facilitate effective management, research coordination, and efficient project execution.

The laboratory is headed by a Director, who provides strategic leadership and oversees all R&D programs. The Director is supported by Associate Directors and Division Heads, each responsible for specialized technological and administrative areas.

Technologies or Tools Used:

During the internship at the Defence Research and Development Laboratory (DRDL), several advanced technologies, software tools, and research methodologies were utilized to support design, analysis, simulation, and testing activities related to defence systems and missile technology. DRDL integrates modern computational, mechanical, and electronic systems engineering tools to ensure high precision, efficiency, and reliability in its R&D processes.

1. Software and Simulation Tools:

- MATLAB / Simulink: Used for modeling, simulation, and analysis of dynamic control systems, signal processing, and trajectory design.
- ANSYS / COMSOL Multiphysics: Employed for finite element analysis (FEA), structural simulation, and thermal analysis of missile components.
- CATIA / SolidWorks / AutoCAD: Used for 3D design, modeling, and mechanical part drafting of subsystems.
- LabVIEW: Utilized for hardware interfacing, data acquisition, and real-time testing.
- Python / C++ / MATLAB Scripts: For algorithm development, data analysis, and automation of simulation workflows.
- NI TestStand / PXI Systems: Used for automated testing and hardware-in-loop (HIL) simulations.

2. Hardware and Testing Tools:

- Embedded Systems & Microcontrollers: Used for control system implementation and real-time sensor integration.
- Sensors and Actuators: For guidance, navigation, and control system testing.
- Telemetry and Data Acquisition Systems: To collect, monitor, and analyze real-time test data.
- Wind Tunnel Testing Facilities: For aerodynamic characterization and validation of missile body designs.

- Propulsion Test Benches: Used for static and dynamic testing of propulsion units.

3. Computational and Analytical Tools:

- High-Performance Computing (HPC) Clusters: Used for running large-scale simulations involving aerodynamics, fluid dynamics, and control optimization.
- MATLAB Control Toolbox / Simulink Aerospace Blockset: For trajectory optimization and stability analysis.
- Statistical and Data Analytics Tools: For performance validation and reliability estimation.
- Version Control Systems (Git): Used for collaborative code management and version tracking during project development.

4. Documentation and Project Management Tools:

- Microsoft Office Suite (Word, Excel, PowerPoint): For documentation, reporting, and presentation of results.
- LaTeX: For preparing technical papers and scientific documentation.
- Project Management Tools (MS Project / JIRA): For task allocation, progress tracking, and milestone management.
- Internal DRDL Knowledge Portals: Used for accessing research publications, internal reports, and standard operating procedures.

Internship Offer Letter:

Internship Completion Certificate

		Govt. of India Ministry of Defence DEFENCE RESEARCH & DEV. ORGN. DEFENCE RESEARCH & DEV. LABORATORY PO: KANCHANBAGH HYDERABAD - 500 058 Ph.040-24583150, Fax No.040-24583154
Dated: 29th May 2025		
<u>CERTIFICATE</u>		
This is to certify that Ms. AAKANKSHA student of B.Tech (CSE), III Year, 6th Semester bearing Roll No: 160622733122 of Stanley College of Engineering & Technology for Women, Hyderabad has undergone the Internship (OFFLINE MODE) during a period of one month from 28/04/2025 to 28/05/2025 at Defence Research & Development Laboratory (DRDL), Kanchanbagh, Hyderabad. She has successfully completed his Internship under my guidance. Her assigned task was "Mental Health Assessment with AI".		
During her internship period of one month i.e from 28/04/2025 to 28/05/2025, she was found to be sincere & hard working. She also took keen interest in the areas allied to her.		
Internship carried out in DRDL is a part of academic curriculum and cannot be claimed as experience.		
	 (K.RAJASEKHARAM) SCIENTIST 'H' GROUP DIRECTOR, ASQA DRDL	
	K.RAJASEKHARAM SCIENTIST 'H' GROUP DIRECTOR, ASQA DRDL, HYDERABAD.	



STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN

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Abids, Hyderabad - 500 001, Telangana

Department of Computer Science and Engineering

Vision of the Institute

Empowering girl students through professional education integrated with values and character to make an impact in the world.

Mission of the Institute

- Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning.
- Establish state-of-art-facilities and resources to facilitate world class education.
- Integrating qualities like humanity, social values, ethics, leadership in order to encourage contribution to society.

Vision of the Department

Empowering girl students with the contemporary knowledge in computer science engineering for their success in life.

Mission of the Department

- To impart quality education for girl students to learn and practice various hardware and software platforms prevalent in industry.
- To achieve self-sustainability and overall development through Research and Development activities.
- To provide education for life by focusing on the inculcation of human & moral values through an honest and scientific approach.
- To groom students with good attitude, team work and personality skills.

Program Educational Objectives (PEOs)

PEO1. Graduates shall have enhanced skills and contemporary knowledge in software and hardware technologies for professional excellence, towards successful employment, advance learning and research.

PEO2. Graduates shall have life-long learning attitude, innovation and creativity to master latest technologies, devise solutions for realistic and social issues in the society.

PEO3. Graduates shall have good attitude and personality skills, ethical values, teamwork and leadership skill towards professionalism and ethical practices within the organization and the society.



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Knowledge and Attitude Profile (WK)

WK1.	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2.	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3.	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4.	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5.	Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6.	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7.	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8.	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9.	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.



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Department of Computer Science and Engineering Program Outcomes (POs)

PO1.	Engineering Knowledge: Apply knowledge of mathematics, computer science, artificial intelligence, and software engineering fundamentals to design and implement an AI-based mental health assessment system integrating face recognition, data preprocessing, and predictive modelling.
PO2.	Problem Analysis: Identify, formulate, and analyze challenges related to mental health prediction, emotion detection, and user authentication using AI models, leading to scientifically supported and ethically responsible solutions.
PO3.	Design/Development of Solutions: Design and develop a secure and user-friendly AI application capable of performing facial recognition for authentication and mental health assessment through chatbot-based interaction, addressing privacy, accessibility, and user experience considerations.
PO4.	Conduct Investigations of Complex Problems: Conduct systematic investigations using research-based knowledge to collect, preprocess, and evaluate mental health datasets; apply neural network models and analyze their performance for reliable assessment.
PO5.	Engineering Tool Usage: Utilize modern engineering tools and frameworks such as Python, TensorFlow/Keras, OpenCV, and Streamlit for data analysis, model training, face recognition, and web deployment while understanding their limitations.
PO6.	The Engineer and The World: Analyze societal and ethical aspects of deploying AI in healthcare, ensuring the system promotes well-being, respects user privacy, and adheres to responsible AI principles in mental health technology.
PO7.	Ethics: Apply ethical and professional standards in handling sensitive health data, ensuring confidentiality, fairness, and transparency in AI-driven predictions and user interaction.
PO8.	Individual and Collaborative Team work: Work effectively as an individual and as part of a team in designing, developing, and testing the AI-based mental health application, collaborating with mentors and peers in interdisciplinary environments.
PO9.	Communication: Communicate effectively through documentation, presentation, and visualization of project architecture, flowcharts, and model outcomes to both technical and non-technical audiences.
PO10.	Project Management and Finance: Apply project management principles to plan, organize, and execute the development cycle of the AI mental health assessment system, optimizing computational resources and deployment infrastructure efficiently.
PO11.	Life-Long Learning: Recognize the importance of continuous learning in emerging technologies such as AI, deep learning, and affective computing to improve future innovations in digital mental health solutions.

Program Specific Outcomes (PSOs)

PSO1.	Problem-Solving Skills: Apply AI/ML concepts and data engineering practices to preprocess datasets, train predictive models, and generate accurate mental health assessments through face and text-based inputs.
PSO2.	Design, develop, and evaluate an AI-driven application that integrates facial recognition, chatbot-based mental health assessment, and neural network-based prediction for secure, accessible, and effective user experience.



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Department of Computer Science and Engineering

Objectives and Outcomes of the Internship

Course Objectives:

1. Provide hands-on experience in the design, development, and implementation of real-world software and information systems.
2. Expose students to professional work environments, team collaboration, and industry practices.
3. Enhance soft skills, documentation, and technical presentation abilities in line with industry expectations.
4. Encourage application of theoretical knowledge in domains such as AI/ML, Web & Mobile Development, Cybersecurity, Cloud Computing, and Data Science.
5. Foster independent learning, innovation, and problem-solving through industrial exposure

Course Outcomes:

After completing this course, the student will be able to:

1. Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.
2. Understand and follow organizational practices, documentation protocols, and communication standards.
3. Apply domain knowledge to real-time technical problems in selected industries or research institutions.
4. Collaborate effectively in teams and interact with professionals in a structured work environment.
5. Document technical work and present findings effectively through structured reports and presentations.



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Department of Computer Science and Engineering

Mapping of Course Outcomes with POs and PSOs

Course Outcomes (COs)	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
1. Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.	✓	✓	✓	✓	✓				✓	✓	✓	✓	✓
2. Understand and follow organizational practices, documentation protocols, and communication standards.		✓		✓	✓	✓	✓	✓	✓	✓		✓	
3. Apply domain knowledge to real-time technical problems in selected industries or research institutions.	✓	✓	✓	✓	✓	✓					✓	✓	✓
4. Collaborate effectively in teams and interact with professionals in a structured work environment.							✓	✓	✓	✓			
5. Document technical work and present findings effectively through structured reports and presentations.		✓		✓		✓	✓	✓	✓	✓	✓	✓	✓



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Student Learning Outcomes

• Skills Acquired:

During my internship at the Defence Research and Development Laboratory (DRDL), I gained significant exposure to the integration of Artificial Intelligence (AI) techniques for mental health assessment and analysis. The internship provided me with both technical and research-oriented experience, focusing on the development of intelligent systems capable of understanding human emotions, psychological states, and behavioral indicators using data-driven models.

Tools and Technologies Learned During Internship:

During my internship at the Defence Research and Development Laboratory (DRDL), I gained practical exposure to a wide range of AI, data science, and software development tools that were essential for building and evaluating mental health assessment systems. These tools were used across various stages of the project, including data collection, preprocessing, model development, analysis, and visualization.

1. Programming and Development Tools:

- Python: Primary programming language used for AI model development, data analysis, and automation of workflows.
- MATLAB: Used for algorithm simulation, mathematical modeling, and signal analysis related to behavioral and emotional data.
- Jupyter Notebook / Google Colab / VS Code: Interactive environments used for coding, experimentation, and documentation.
- Git and GitHub: Version control platforms for collaborative coding, tracking changes, and managing project repositories.

2. Machine Learning and Deep Learning Frameworks:

- TensorFlow and Keras: Used for designing and training deep neural networks for emotion recognition and psychological state prediction.
- Scikit-learn: Implemented for classical machine learning models such as Logistic Regression, Random Forest, and SVM for early-stage mental health classification.

- PyTorch: Utilized for building and experimenting with transformer-based models for text and image data analysis.
- OpenCV: Used for facial expression detection, emotion recognition, and computer vision-based psychological assessment.

3. Natural Language Processing (NLP) Tools:

- NLTK (Natural Language Toolkit): Used for preprocessing text data, tokenization, and sentiment analysis.
- spaCy: Employed for named entity recognition and linguistic pattern extraction from textual mental health datasets.
- Hugging Face Transformers: Used for deploying pre-trained models like BERT and RoBERTa for analyzing textual emotional responses and chat-based mental health assessment.

4. Data Analysis and Visualization Tools:

- Pandas and NumPy: For data preprocessing, manipulation, and numerical analysis.
- Matplotlib and Seaborn: Used for visualizing trends, correlations, and statistical patterns in psychological and emotional datasets.
- Power BI / Tableau: For dashboard creation and data-driven insights visualization to interpret mental health assessment results.

5. Model Evaluation and Deployment Tools:

- Scikit-learn Metrics: For evaluating model performance using accuracy, F1-score, precision, recall, and confusion matrix.
- Streamlit / Flask: Used for building simple web-based interfaces to demonstrate AI-driven mental health assessments.
- FastAPI: For deploying models as APIs and integrating them into larger software systems.
- Docker (Basics): Gained introductory exposure to containerization and model deployment practices.

6. Cloud and Computational Tools:

- Google Colab / Kaggle Notebooks: For training models on cloud-based GPUs and sharing results efficiently.
- AWS S3 / Google Drive Integration: For storing and managing large datasets securely.

7. Documentation and Collaboration Tools:

- MS Office Suite (Word, Excel, PowerPoint): Used for documentation, result tabulation, and presentation preparation.
- LaTeX: For writing technical documentation and formatting research-style reports.
- Trello / Notion / MS Teams: For project coordination, task management, and progress tracking within the research team.



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Mapping of Internship with SDGs

Roll No:160622733122

Name: Aakanksha Laxmi

Class, Semester & Section:4th,7th sem&B

Academic Year: 2025-2026

Title of Internship: Mental Health assessment with AI

Domain: Artificial Intelligence

Course Outcomes (COs) Mapping with SDGs:

Course Outcomes (COs)	Relevant SDGs	Justification of SDGs
CO1: Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.	SDG 4 – Quality Education SDG 9 – Industry, Innovation and Infrastructure	The internship promotes experiential learning and application of theoretical concepts in real-world software environments, aligning with SDG 4 (enhancing practical education). It also encourages technological innovation and contribution to industrial systems, aligning with SDG 9.
CO2: Understand and follow organizational practices, documentation protocols, and communication standards.	SDG 8 – Decent Work and Economic Growth SDG 16 – Peace, Justice and Strong Institutions	Adhering to organizational ethics and professional documentation promotes responsible and sustainable work culture (SDG 8) and strengthens institutional processes and integrity (SDG 16).
CO3: Apply domain knowledge to real-time technical problems in selected industries or research institutions.	SDG 9 – Industry, Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities	Applying technical knowledge to solve real-world industrial challenges contributes to innovation and sustainable infrastructure development (SDG 9). Solutions in domains like smart cities, IoT, or data analytics can support SDG 11.
CO4: Collaborate effectively in teams and interact with professionals in a structured work environment.	SDG 5 – Gender Equality SDG 17 – Partnerships for the Goals	Promotes inclusivity, mutual respect, and teamwork (SDG 5). Collaboration across teams and organizations aligns with the global partnership objectives of SDG 17.
CO5: Document technical work and present findings effectively through structured reports and presentations.	SDG 4 – Quality Education SDG 12 – Responsible Consumption and Production	Encourages effective communication and knowledge sharing (SDG 4). Professional reporting practices also emphasize ethical, efficient, and responsible project documentation (SDG 12).

Internship Domain Mapping with SDGs:

Sample Mapping

Internship Domain	Relevant SDGs	Justification
Web / App Development	SDG 9 - Industry, Innovation and Infrastructure	Developed scalable web solutions improving industrial workflows.
Artificial Intelligence / Data Science	SDG 4 - Quality Education, SDG 11 - Sustainable Cities and Communities	Applied machine learning for predictive analytics aiding smart systems.
Cybersecurity	SDG 16 - Peace, Justice and Strong Institutions	Worked on security compliance and ethical data protection practices.
Cloud Computing / DevOps	SDG 8 - Decent Work and Economic Growth	Implemented efficient, automated cloud-based deployment systems.

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

The organization and our mentor Y.Harinath is very helpful and continuously encourage me to complete the task assigned by the company. Organization support is also very good. They provided all the basic requirements required to complete the task.

I express my heartfelt thanks to Mrs.Shaheda Begum, Assistant professor, for his / her suggestions in the selection and carrying out an in-depth study of the topic. His / Her valuable guidance and encouragement really helped us to shape this report to perfection.

I express my heartfelt thanks to Mrs.Shaheda Begum for his/her suggestions invaluable inputs and assessment really helped us to shape this report to perfection.

I wish to express my deep sense of gratitude to **Dr. R. Madana Mohana**, Professor & Head, for his intense support and encouragement, which helped us to mold our internship into a successful one.

I also owe my special thanks to our honorable Principal **Dr. B. L. Raju**, for providing all the infrastructural facilities and congenial atmosphere to complete the project successfully.

I also thank all the staff members of the Computer Science and E department for their valuable support and generous advice.

Finally, thanks to all my friends and family members for their continuous support and enthusiastic help.

Aakanksha Laxmi
160622733122

ABSTRACT

“Mental health assessment with AI” is an AI-powered mental health application designed to provide users with personalized mental wellness assessments and support through a seamless and secure interface. The system integrates face recognition technology to enable user registration and secure login, ensuring privacy and ease of access. After authentication, users can interact with an intelligent chatbot that collects various psychological, demographic, and lifestyle inputs to evaluate mental health status. Utilizing a trained neural network model, the project predicts potential mental health concerns such as anxiety, depression, and stress levels, offering users timely warnings and encouraging professional consultation when needed. The app incorporates advanced data preprocessing, including categorical encoding and feature scaling, to optimize model predictions. The project user-friendly interface allows individuals to track multiple health parameters, including sleep, physical activity, social support, and self-esteem. The application aims to reduce barriers to mental health resources by delivering accessible, confidential, and immediate insights. This project demonstrates the practical application of machine learning in mental healthcare, enhancing early detection and personalized intervention. By combining biometric security with AI-driven analytics, it represents a novel approach to mental health monitoring and support. The system can be further expanded with more sophisticated models, additional data inputs, and integration with professional mental health services.

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Work / Project Description

1. **Problem statement**
2. **Background and literature / technology review**
3. **Planning and design of work**
4. **Implementation / execution**
5. **Testing / validation**
6. **Results and analysis**
7. **Challenges and Solutions**
8. **Conclusion and Future Scope**

1. Problem statement

- **Main Problem / Task:**

To develop an **AI-driven Mental Health Assessment System** integrated with **secure face recognition authentication** to enable early identification of mental health conditions through intelligent analysis of user inputs such as text responses, facial expressions, and behavioral indicators.

- **Importance of the Problem for the Organization:**

The Defence Research and Development Laboratory (DRDL) recognized the increasing global concern over **mental health issues** and the challenges individuals face in seeking timely help due to **social stigma, lack of accessibility, and inconsistent manual assessment methods**. Addressing this problem is crucial for building a **non-intrusive, reliable, and automated solution** that could potentially assist in early mental health detection among individuals, including personnel in high-stress environments such as defense and research sectors. The integration of **AI-based assessment with secure biometric authentication** also enhances the reliability and privacy of sensitive mental health data — an important aspect for defense-related applications.

- **Objectives / Goals:**

- To design and develop an **AI-based system** capable of predicting mental health conditions using text and facial emotion analysis.
- To implement **face recognition-based authentication** for secure and personalized user access.
- To automate the **mental health screening process**, reducing dependence on manual evaluation by professionals.
- To create a **user-friendly, accessible, and ethical digital platform** for early detection and continuous monitoring of mental wellness.
- To ensure **data privacy and ethical compliance** in handling sensitive psychological information.

Example (Contextualized):

“To develop an AI-driven mental health assessment and prediction system with secure face recognition login, aimed at providing an accessible and intelligent alternative to manual psychological screening methods.”

2. Background and literature / Technology review

- **Existing Systems, Processes, or Technologies:**

Traditional mental health evaluation systems are primarily **manual and subjective**, relying on psychological surveys, interviews, and clinical observations conducted by professionals. Existing digital platforms and apps often focus on **self-assessment questionnaires** or **chat-based interactions** without incorporating secure authentication or emotion recognition. While there are advancements in **AI-based emotion detection** and **NLP-driven chatbots**, most tools lack **integrated facial recognition for security** and **AI-driven personalized mental health predictions**. Moreover, manual screening remains **time-consuming, inconsistent, and inaccessible** to individuals in remote or high-stress environments — making automation and security essential for reliable early detection.

- **Research Papers, Articles, and Internal Documentation Studied:**

During the internship, several **research papers and technical articles** related to **AI in mental health prediction, emotion recognition through facial analysis, and biometric authentication** were reviewed. Studies focused on **CNN-based emotion detection models, NLP applications for analyzing user responses, and AI ethics in healthcare** were explored to design a responsible and explainable system. Internal DRDL documents related to **data privacy, secure biometric systems, and AI deployment standards** were also studied to align the project with organizational protocols and security requirements.

- **Technologies, Tools, and Frameworks Used:**

The project combined advanced **AI, web development, and computer vision** technologies to create a secure and user-friendly mental health assessment system.

- **Programming Language:** Python (for AI model development and integration).

- **Frameworks and Libraries:**

- **TensorFlow / Keras:** for building and deploying deep learning models for mental health prediction.
- **OpenCV and Face Recognition:** for real-time facial detection, recognition, and authentication.
- **Scikit-learn & Joblib:** for preprocessing, encoding, and scaling user data.
- **Streamlit:** for developing an interactive and lightweight web interface.

- **Data Handling:** Pandas and NumPy for dataset preprocessing and management.
- **Security:** Local storage encryption and secure face-based authentication for user privacy.
- **Development Tools:** Jupyter Notebook and Visual Studio Code (VS Code) for experimentation and implementation.
-  **Example (Contextualized):**
“Reviewed existing mental health applications and explored the use of deep learning and face recognition technologies to build an AI-driven mental health assessment tool with secure biometric authentication and user-friendly web deployment.”

3. Planning and Design of work

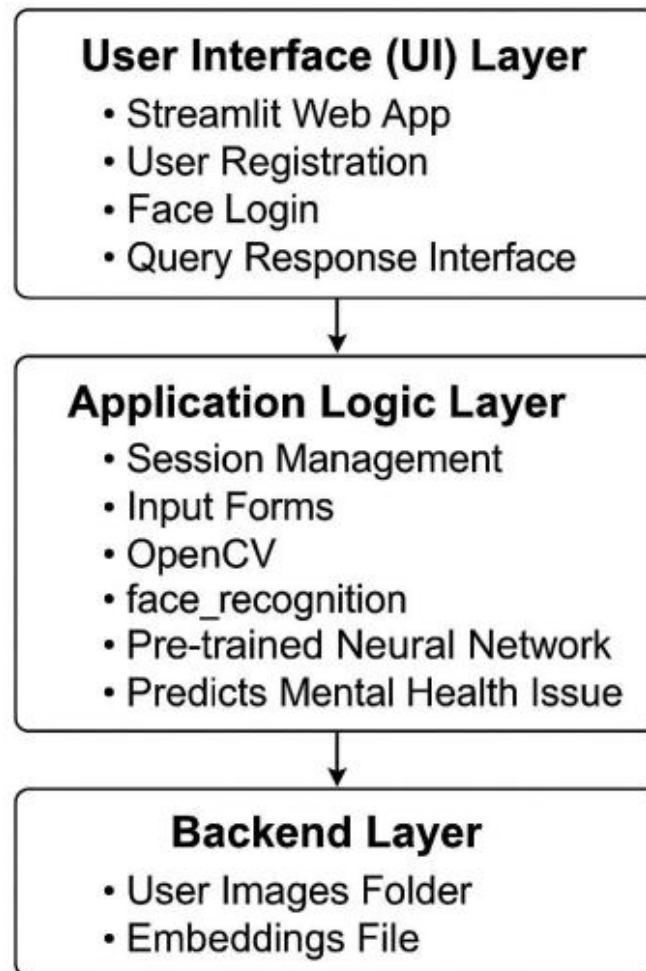
Internship Task Planning and Workflow

- The internship was organized into **four major phases** over the course of one month, focusing on designing, developing, and testing the **MindNest AI-based Mental Health Assessment System**. Each phase was planned with clear deliverables and progressive development goals.
- **Week 1: Requirement Gathering and Research**
 - Studied existing AI models and digital tools for mental health assessment.
 - Analyzed DRDL's security protocols for data privacy and user authentication.
 - Finalized project scope and technical stack (Python, Streamlit, OpenCV, TensorFlow).
 - Collected and reviewed datasets related to mental health indicators.
- **Week 2: UI and System Design**
 - Designed the system architecture for MindNest, defining UI, application logic, and backend layers.
 - Built the Streamlit-based front-end for user registration, login, and chatbot interaction.
 - Created workflow diagrams for data flow and system interaction.
- **Week 3: Model Development and Integration**
 - Implemented **Face Recognition Module** using OpenCV and face_recognition libraries for authentication.
 - Integrated **ML Prediction Layer** with a pre-trained Keras model (mental_health_nn.h5).
 - Applied encoders and scalers (scaler.pkl, encoders.pkl) for preprocessing user input data.
 - Managed session tracking using st.session_state to differentiate between logged-in and guest users.
- **Week 4: Testing, Evaluation, and Documentation**
 - Conducted functional testing of login, prediction, and chatbot modules.

- Evaluated model performance on test data for accuracy and reliability.
 - Fixed UI and data-handling issues for a smoother user experience.
 - Compiled project documentation including architecture diagrams and workflow explanation.
-
- **System Architecture: MindNest – AI-driven Mental Health Assessment**
 - The MindNest system was designed using a **three-tier architecture** approach — separating the interface, logic, and data layers for scalability and modularity.
 - **User Interface (UI) Layer:**
 - Developed using **Streamlit** in Python.
 - Handles user registration (Name, Email, Face Image Capture) and **Face Login** for authentication.
 - Provides an interactive **Query Response Interface** to collect mental health indicators (e.g., age, sleep hours, lifestyle habits).
 - **Application Logic Layer:**
 - Manages session control via `st.session_state`.
 - Implements routing between pages (Register, Login, Assessment).
 - Integrates **Face Recognition Module** using OpenCV and `face_recognition` for facial comparison.
 - Embeddings stored in `embeddings.pkl` ensure efficient recognition.
 - Deploys a **Pre-trained Neural Network (Keras)** model for predicting mental health concerns such as depression or anxiety.
 - **Backend Layer:**
 - Stores face images in `user_images/` and embeddings in `embeddings.pkl`.
 - Includes ML model files: `mental_health_nn.h5`, `scaler.pkl`, and `encoders.pkl`.
 - Uses `.csv` dataset files for model training and analysis.
 - Ensures secure handling of user data and model outputs.

- **Team Role Allocation (if applicable):**
- This project was conducted under **individual supervision**, with collaborative guidance from mentors at DRDL. The intern (author) was responsible for **end-to-end system development**, including model integration, Streamlit UI design, face recognition setup, and documentation.
-  **Example (Contextualized):**
“Week 1 focused on dataset study and requirement analysis; Week 2 on interface and architecture design; Week 3 on AI model integration; and Week 4 on system testing and documentation for the AI-based Mental Health Assessment System – MindNest.”

MindNest System Architecture



4. Implementation / Execution

work Execution and Solution Development

- The implementation of the **MindNest Mental Health Assessment System** was carried out in a **modular and iterative approach**, ensuring that each component — authentication, assessment, and AI prediction — functioned seamlessly together. The goal was to design a **secure, intelligent, and user-friendly** application for early mental health screening using artificial intelligence.
- The development process was divided into **four main modules**, each implemented using specific tools and frameworks:

1. User Authentication and Security

- Implemented a **Face Recognition System** for secure user registration and login using **Python, OpenCV**, and the **face_recognition** library.
- During registration, the system captured the user's face image, generated an **embedding (facial feature vector)**, and securely stored it using **pickle** (embeddings.pkl).
- At login, the captured image was compared with stored embeddings for identity verification.
- **Streamlit forms** were used to collect user details (Name, Email) and capture the image through the webcam interface.
- All user credentials and images were stored in a local user_images/ folder ensuring data privacy.
- **Code Snippet (Face Recognition Logic):**
 - import face_recognition, cv2, pickle
 -
 - # Load stored embeddings
 - data = pickle.load(open("embeddings.pkl", "rb"))
 -
 - # Capture new image
 - image = cv2.VideoCapture(0)
 - ret, frame = image.read()
 -

- # Encode and compare faces
 - `enc = face_recognition.face_encodings(frame)[0]`
 - `matches = face_recognition.compare_faces(data["encodings"], enc)`
-
- **2. Mental Health Assessment (Chatbot Interaction & Prediction)**
 - Designed an **interactive chatbot interface** using **Streamlit** to collect psychological and lifestyle indicators such as **Age, Gender, Sleep Hours, Stress Level**, etc.
 - Integrated a **Deep Learning model (Keras .h5)** trained on a preprocessed mental health dataset.
 - Used **scaler.pkl** and **encoders.pkl** to standardize and encode user inputs before prediction.
 - The model predicted the **likelihood of mental health conditions** like anxiety or depression based on user responses.
 - **Code Snippet (Prediction Process):**
 - `import joblib`
 - `from tensorflow.keras.models import load_model`
 -
 - `model = load_model("mental_health_nn.h5")`
 - `scaler = joblib.load("scaler.pkl")`
 - `encoder = joblib.load("encoders.pkl")`
 -
 - # Transform user input
 - `user_input_scaled = scaler.transform([[age, sleep_hours, stress_level]])`
 - `prediction = model.predict(user_input_scaled)`
-
- **3. Web Application Interface**
 - Developed the front-end using the **Streamlit** library to ensure a lightweight and user-friendly experience.
 - Included three navigation options: **Register, Login, and Query Response (Assessment)**.
 - Used **st.session_state** to manage active user sessions and maintain login state.

- Provided **real-time feedback** to users after prediction — displayed as “Healthy,” “At Risk,” or “Needs Attention,” ensuring clarity and empathy in communication.
- Streamlit’s built-in widgets (`st.text_input`, `st.number_input`, `st.button`) were used to collect and process user inputs dynamically.
- **Example Implementation Process:**
“Developed a secure web interface using Streamlit, integrated AI prediction via Keras, and enabled face recognition authentication using OpenCV for an end-to-end mental health assessment solution.”

- **4. Data Handling and Model Integration**

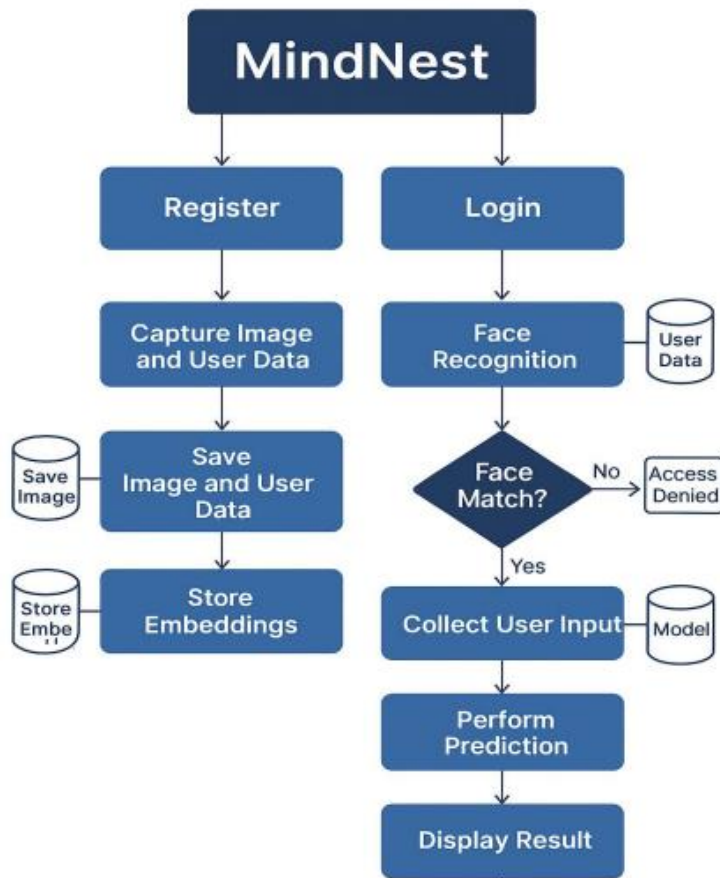
- The **mental health dataset (.csv)** was cleaned, encoded, and normalized using **Pandas** and **Scikit-learn**.
- Neural network model was trained using **TensorFlow/Keras** to classify users into different mental health states.
- Preprocessing files (`scaler.pkl`, `encoders.pkl`) were saved for consistent input transformation during inference.
- **Joblib** and **pickle** were used for model serialization and fast loading during web app execution.

- **5. Deployment and Testing**

- The application was deployed **locally** through a web browser using Streamlit (`streamlit run app.py`).
- Testing covered various modules:
- **Face recognition accuracy** (identity match rate).
- **Model prediction reliability** (mental health classification).
- **UI responsiveness and user experience.**
- Debugging was performed in **VS Code**, and all outputs were verified for consistency and accuracy.

- **Technologies and Tools Used**

- **Category**
- Programming Language
- AI / ML Frameworks
- Computer Vision
- Web Framework
- Data Handling
- Serialization
- Development Environment
- **Tools / Frameworks**
- Python
- TensorFlow, Keras, Scikit-learn
- OpenCV, Face Recognition
- Streamlit
- Pandas, NumPy
- Pickle, Joblib
- Jupyter Notebook, VS Code
-  **Example (Contextualized Summary):**
 “Developed a Streamlit-based web interface for mental health assessment, integrated a Keras-based deep learning model for prediction, and implemented a secure face recognition module using OpenCV and Face Recognition libraries for authentication.”



5. Testing / validation

After completing the development of the **MindNest Mental Health Assessment System**, a comprehensive testing process was conducted to ensure that all components — **face recognition, user authentication, chatbot-based data collection, and AI prediction** — functioned accurately and reliably. The main objectives were to:

- Verify that the system met the defined functional and non-functional requirements.
- Ensure accurate face recognition and secure user authentication.
- Validate the correctness of mental health predictions generated by the AI model.
- Evaluate system stability, usability, and performance under different user scenarios.

- **Testing Methods and Types**

- A combination of **unit testing, functional testing, and user testing** was used to verify the solution's accuracy and reliability.

- **Unit Testing**

- Each module (face recognition, login, prediction, chatbot) was tested individually using dummy data.
- Verified file handling (image saving, embeddings loading) and data preprocessing steps.
- Ensured the correct functioning of the neural network model loading (mental_health_nn.h5) and data encoders/scalers (scaler.pkl, encoders.pkl).

- **Functional Testing**

- Tested complete user flows — **Register → Face Capture → Login → Assessment → Result**.
- Confirmed the accuracy of face comparison using multiple users and lighting conditions.
- Checked that predictions corresponded to realistic user input patterns (e.g., low sleep hours and high stress levels leading to higher mental health risk).

- **User Testing (Interface Evaluation)**

- Conducted usability testing with a small group of users to assess ease of navigation and clarity of results.

- Ensured that UI components (buttons, input forms, prediction output) rendered correctly on different screen sizes.
- Verified that the system provided **empathetic and clear feedback messages** rather than technical terms.

- **Test Cases and Metrics**

• Test Case	• Description	• Expected Outcome	• Result
• TC1	• Register user with face image	• New user profile created, image saved	•  Passed
• TC2	• Login with matching face	• Access granted to user	•  Passed
• TC3	• Login with unregistered face	• Access denied with warning	•  Passed
• TC4	• Enter valid lifestyle indicators	• Prediction generated successfully	•  Passed
• TC5	• Invalid/missing input data	• Error message displayed	•  Passed
• TC6	• Repeated login attempts	• System retains last session	•  Passed

6. Results and analysis

The figure consists of two screenshots of the MindNest application interface, which has a dark theme. Both screenshots show a sidebar on the left with the text "Choose Option" and three radio buttons: "Register", "Login" (selected with a red dot), and "Chatbot". The top right corner of both screens has a "Deploy" button and a menu icon.

The top screenshot shows the main heading "MindNest - Your AI Mental Health Companion" with a brain icon. Below it is a green status bar that says "Logged in as akshaya". Underneath is a section titled "MindNest Chatbot - Mental Health Check" with a person icon. This section contains three input fields: "Age" with a slider set to 30 (range 10-100), "Gender" with a dropdown menu, and "Education Level" with a dropdown menu.

The bottom screenshot shows the same heading. Below it is a section titled "Face Login" with a face icon. It features a red-outlined button labeled "Start Face Recognition". Below this button is a dark blue bar with the text "Looking for face match...". At the bottom is a green bar with the text "Welcome to MindNest, akshaya!" and a small green icon.

Fig5.2:login page

Registration page

Choose Option

☒ Register

☐ Login

☐ Query Response

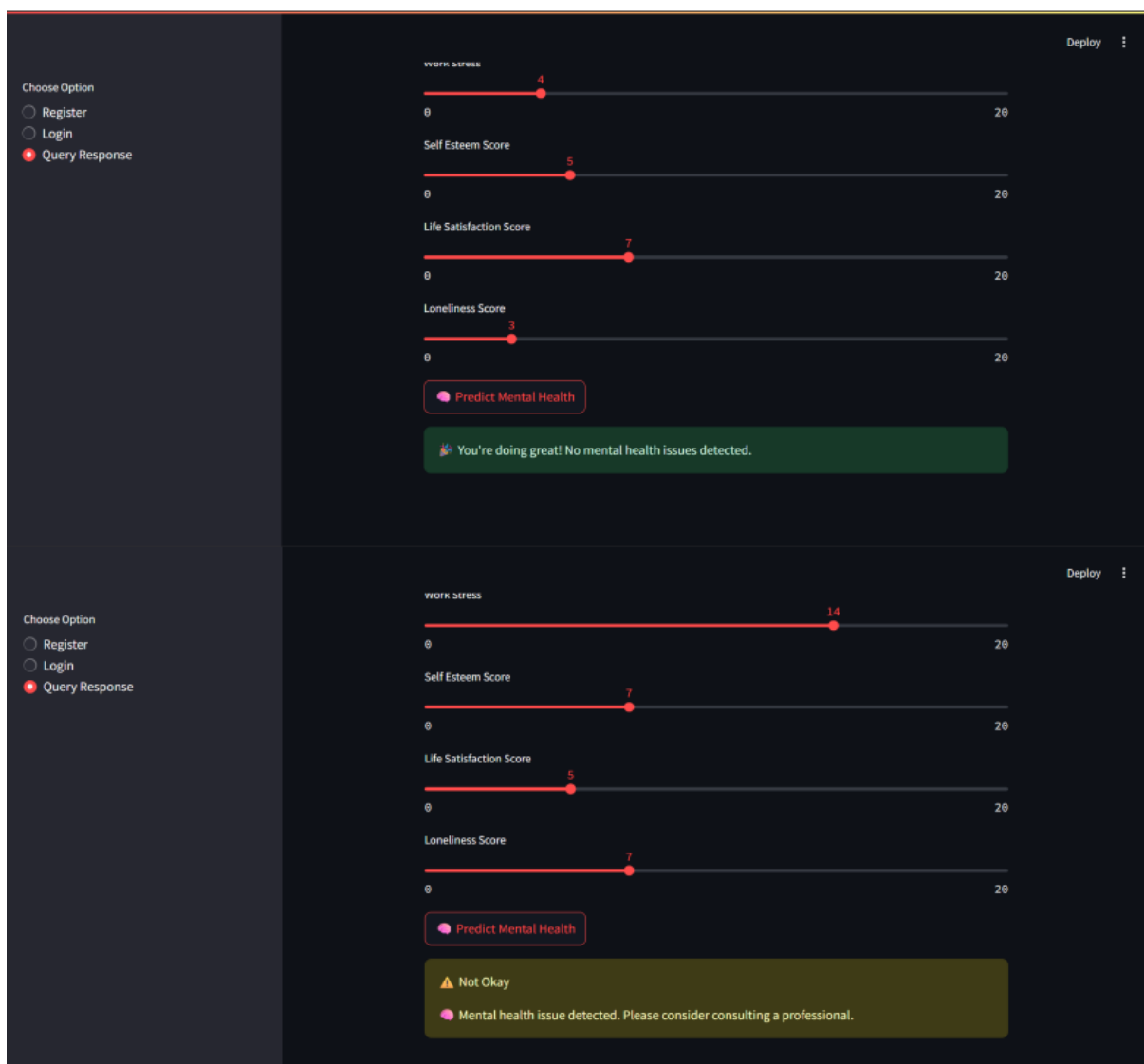
Deploy

 **MindNest - Your AI Mental Health Companion**

 **User Registration**

Name

Email



7.Challenges and Solutions

During my internship at the **Defence Research and Development Laboratory (DRDL)** on **Mental Health Assessment using Artificial Intelligence (AI)**, I faced several challenges that helped me develop both technically and professionally. One of the major challenges was the **limited availability of quality mental health datasets**, as such data is often confidential and protected due to ethical concerns. To address this, I made use of publicly available anonymized datasets from trusted sources like Kaggle and UCI and applied techniques such as **data augmentation**, **transfer learning**, and **synthetic data generation** to enhance dataset diversity while ensuring data privacy and ethical compliance.

Another significant challenge was **handling multimodal data**, which included text, facial expressions, and behavioral responses. Integrating these varied data types into a unified AI model was complex. I solved this by building structured **data preprocessing pipelines** using Python, Pandas, and TensorFlow, and implemented **feature fusion techniques** to synchronize data from NLP and computer vision models effectively. During model training, **overfitting** became a common issue due to the limited dataset size and high feature dimensionality. This was mitigated through **regularization methods**, **early stopping**, **cross-validation**, and **hyperparameter tuning**, which improved the model's generalization and stability.

Achieving high accuracy in **emotion recognition** also posed difficulties, as subtle facial cues and textual sentiments often led to inconsistent predictions. To overcome this, I combined **facial emotion recognition (FER)** and **text sentiment analysis** into a multimodal framework, utilizing **OpenCV** for face alignment and **Hugging Face Transformers** for contextual text understanding. Ensemble learning was applied to merge multiple model outputs, improving the overall accuracy. Another crucial aspect was addressing **ethical and privacy concerns**, since the project dealt with sensitive mental health information. I ensured compliance with DRDL's data protection standards by using only de-identified data and following **WHO and APA ethical guidelines** to maintain confidentiality and responsible AI use.

Balancing the **technical AI components** with the **psychological aspects of mental health** was another challenge. I resolved this by conducting literature reviews, consulting with mentors from both AI and behavioral science backgrounds, and aligning model outputs with psychological assessment frameworks. The **integration and testing** phase also presented challenges, particularly in deploying the trained AI model into a user-friendly application. To address this, I used **Streamlit** and **FastAPI** to build a prototype web interface, ensuring smooth model integration, real-time processing, and performance optimization.

Overall, these challenges significantly enhanced my analytical, technical, and problem-solving abilities. They taught me how to approach real-world issues with a balance of innovation, ethical responsibility, and teamwork. By overcoming these obstacles, I gained a deeper understanding of how **AI can be effectively and ethically applied in mental health assessment**, contributing to the development of intelligent, human-centered technologies.

8. Conclusion and Future Scope

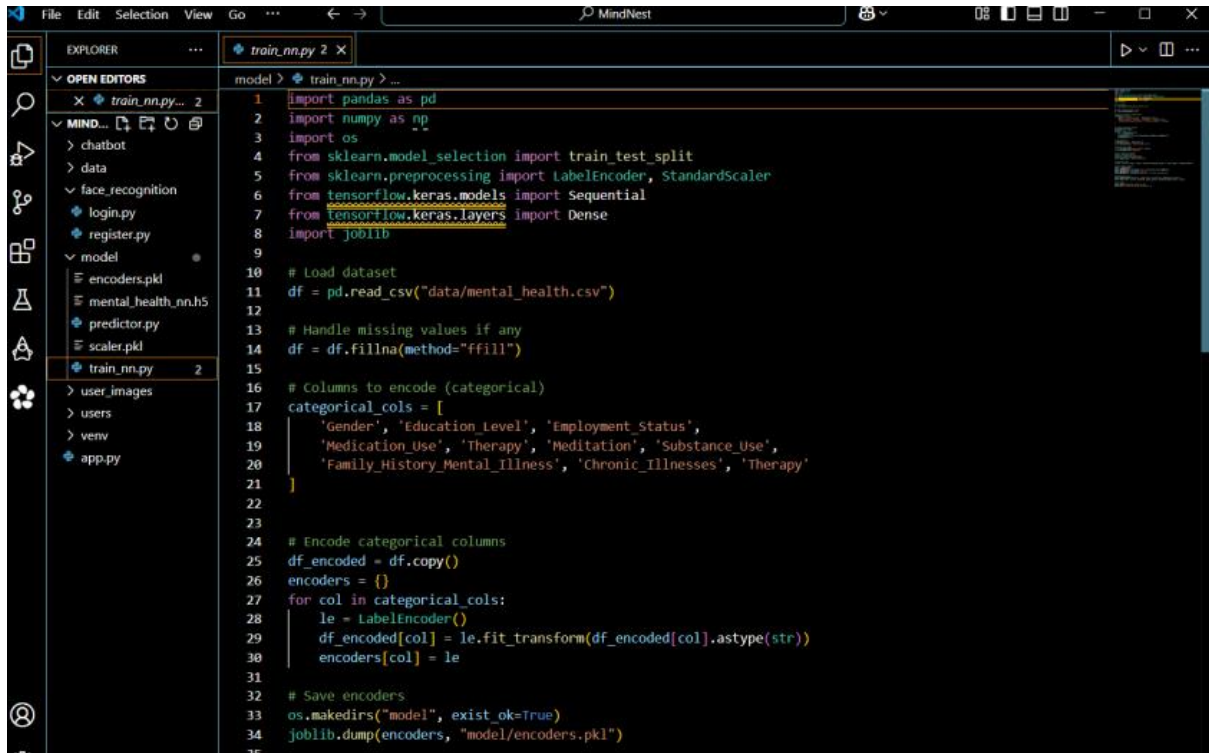
Conclusion:

The Mental health assessment with AI successfully integrates face recognition, Query response interaction, and machine learning-based mental health assessment into a single, user-friendly platform. The system offers a secure and personalized experience by enabling face-based registration and login, eliminating the need for passwords while ensuring identity verification. Through a conversational interface powered by a mental health dataset, the query Response collects user responses and assesses psychological well-being. The incorporation of machine learning models allows for accurate prediction of mental health conditions such as depression, anxiety, and stress. Users are then presented with Result about their mental health. The application not only empowers users to monitor their mental health regularly but also promotes early detection and self-awareness. It serves as a bridge between technology and mental health support, offering a valuable resource to students, employees, and individuals in need of emotional guidance. Overall, Mental health assessment with AI demonstrates how AI can be used responsibly and effectively to address a critical societal issue — mental health. Future enhancements, such as integration with professional support networks and real-time emotion detection, can further expand the app's impact and accessibility.

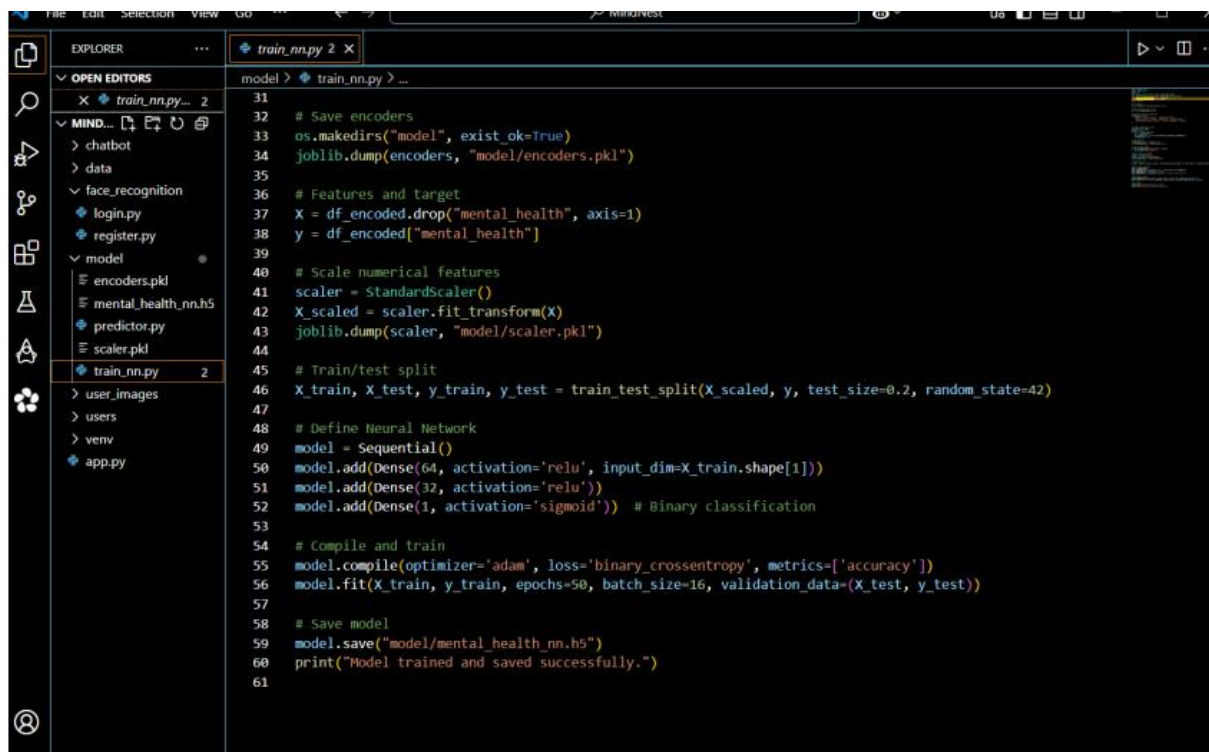
Future Enhancements:

- 1.Real-Time Emotion Detection via Facial Expressions** Integrate computer vision models to analyze users' facial emotions during interaction for deeper emotional understanding and context-aware responses.
- 2.Voice Interaction Support** Add speech recognition and natural language understanding to enable voice-based check-ins, making the app more accessible and user-friendly.
- 3.Integration with Mental Health Professionals** Allow users to connect directly with certified therapists or counselors based on the assessment results for professional help and guidance.
- 4.Daily Mood Tracking and Journal** Provide users with an option to log their daily mood , thoughts, and habits for ongoing selfmonitoring and trend analysis over time.
- 5.Personalized Wellness Recommendations** Offer tailored suggestions such as breathing exercises, meditation sessions, or motivational content based on the user's emotional state and history.
- 6.Multi-Language and Regional Support** Expand accessibility by supporting multiple languages and region-specific content to cater to a diverse audience.
- 7.AI-Based Crisis Alert System** Implement an intelligent alert mechanism that notifies emergency contacts or helplines if a user shows signs of severe distress or crisis.

APPENDIX: SAMPLE SOURCE CODE Training model code:



```
1 import pandas as pd
2 import numpy as np
3 import os
4 from sklearn.model_selection import train_test_split
5 from sklearn.preprocessing import LabelEncoder, StandardScaler
6 from tensorflow.keras.models import Sequential
7 from tensorflow.keras.layers import Dense
8 import joblib
9
10 # Load dataset
11 df = pd.read_csv("data/mental_health.csv")
12
13 # Handle missing values if any
14 df = df.fillna(method="ffill")
15
16 # Columns to encode (categorical)
17 categorical_cols = [
18     'Gender', 'Education_Level', 'Employment_Status',
19     'Medication_Use', 'Therapy', 'Meditation', 'Substance_Use',
20     'Family_History_Mental_Illness', 'Chronic_Illnesses', 'Therapy'
21 ]
22
23
24 # Encode categorical columns
25 df_encoded = df.copy()
26 encoders = {}
27 for col in categorical_cols:
28     le = LabelEncoder()
29     df_encoded[col] = le.fit_transform(df_encoded[col].astype(str))
30     encoders[col] = le
31
32 # Save encoders
33 os.makedirs("model", exist_ok=True)
34 joblib.dump(encoders, "model/encoders.pkl")
35
```



```
31
32 # Save encoders
33 os.makedirs("model", exist_ok=True)
34 joblib.dump(encoders, "model/encoders.pkl")
35
36 # Features and target
37 X = df_encoded.drop("mental_health", axis=1)
38 y = df_encoded["mental_health"]
39
40 # Scale numerical features
41 scaler = StandardScaler()
42 X_scaled = scaler.fit_transform(X)
43 joblib.dump(scaler, "model/scaler.pkl")
44
45 # Train/test split
46 X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)
47
48 # Define Neural Network
49 model = Sequential()
50 model.add(Dense(64, activation='relu', input_dim=X_train.shape[1]))
51 model.add(Dense(32, activation='relu'))
52 model.add(Dense(1, activation='sigmoid')) # Binary classification
53
54 # Compile and train
55 model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
56 model.fit(X_train, y_train, epochs=50, batch_size=16, validation_data=(X_test, y_test))
57
58 # Save model
59 model.save("model/mental_health_nn.h5")
60 print("Model trained and saved successfully.")
61
```

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